

A Density-Continuum Obstruction for BCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving the commutative $C(K)$ criterion and the von Neumann algebra criterion, it asks for a noncommutative C^* -algebra version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Theorem 1. *Let X be a complex Banach space. If X has BCP, then*

$$|X| \leq |\mathbb{C}^{\mathbb{N}}| = \mathfrak{c}.$$

In particular $\text{dens}(X) \leq \mathfrak{c}$.

Corollary 1. *Let A be a C^* -algebra. If $\text{dens}(A) > \mathfrak{c}$, then A fails BCP.*

Idea of the Proof

A BCP cover has countably many centres c_n and radii $r_n < \|c_n\|$. Choose one norming functional f_n for each nonzero centre. If the functionals failed to separate points, a nonzero common-kernel vector u could be normalized. Since $f_n(u) = 0$, every centre would be at distance at least $\|c_n\|$ from u , so u would escape all balls in the cover. Therefore the chosen norming functionals separate points, and X embeds linearly into $\mathbb{C}^{\mathbb{N}}$. This forces cardinality at most continuum.

Proof

Proof of Theorem 1. Suppose X has BCP. Use the equivalent open-ball formulation. Then there are centres $c_n \in X$ and radii $r_n > 0$ such that

$$S_X \subseteq \bigcup_{n=1}^{\infty} B^\circ(c_n, r_n), \quad 0 \notin B^\circ(c_n, r_n)$$

for every n . Thus $r_n \leq \|c_n\|$, and no centre relevant to the cover is zero.

For each n , choose $f_n \in X^*$ with

$$\|f_n\| = 1, \quad f_n(c_n) = \|c_n\|,$$

using Hahn–Banach.

We claim that $\{f_n : n \in \mathbb{N}\}$ separates points of X . If not, there is a nonzero $v \in X$ such that $f_n(v) = 0$ for every n . Put $u = v/\|v\| \in S_X$. Then, for every n ,

$$\|u - c_n\| \geq |f_n(u - c_n)| = | - \|c_n\| | = \|c_n\| \geq r_n.$$

Thus u belongs to none of the open covering balls, a contradiction.

Therefore the linear map

$$T : X \rightarrow \mathbb{C}^{\mathbb{N}}, \quad T(x) = (f_n(x))_{n=1}^{\infty},$$

is injective. Since $|\mathbb{C}^{\mathbb{N}}| = \mathfrak{c}$, we get $|X| \leq \mathfrak{c}$. The density character of a metric space is bounded above by its cardinality, so $\text{dens}(X) \leq \mathfrak{c}$. $\square$

Proof of Corollary 1. A C^* -algebra is a complex Banach space under its C^* -norm. If it had BCP, Theorem 1 would give density at most $\mathfrak{c}$. Hence density strictly larger than $\mathfrak{c}$ forces failure of BCP. $\square$

Scope for the C^* -Classification

This obstruction is independent of primitive ideal theory. It applies to all Banach spaces, hence to all C^* -algebras. In the C^* -classification program for arXiv:2311.14257, it removes all algebras of density greater than the continuum from the positive side.

It does not settle the remaining density- $\mathfrak{c}$ simple nonseparable case. The countable product $\ell_\infty(\mathcal{O}_2)$ has density $\mathfrak{c}$ and UBCP by the product dichotomy packet, so the cardinal bound is sharp even among nonseparable non-Type-I C^* -algebras.

Novelty and Search Bounds

The local run indexes were searched for density, continuum, algebraic dimension, countable separating families, and BCP. Nearby Banach-space cardinality packets were found, but no prior lane-19 packet recording this BCP density-continuum obstruction for arXiv:2311.14257 was found. Web searches during the surrounding classification push focused on BCP for C^* -algebras, separable subalgebras, relative commutants, and simple nonseparable C^* -algebras.

Human Review Focus

Reviewers should check:

1. the off-origin radius convention for BCP balls;
2. the use of Hahn–Banach norming functionals;
3. the argument that a nonzero common-kernel vector escapes all balls.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Run `fa_banach_001`, lane 19, *When the Countable-Capture BCP Obstruction Exists*, solution packet `2311.14257_countable_capture_characterization`, 2026.
- [3] Run `fa_banach_001`, lane 19, *A Product Dichotomy for Separable C^* -Algebras and BCP*, solution packet `2311.14257_countable_products_separable_cstar_ubcp`, 2026.